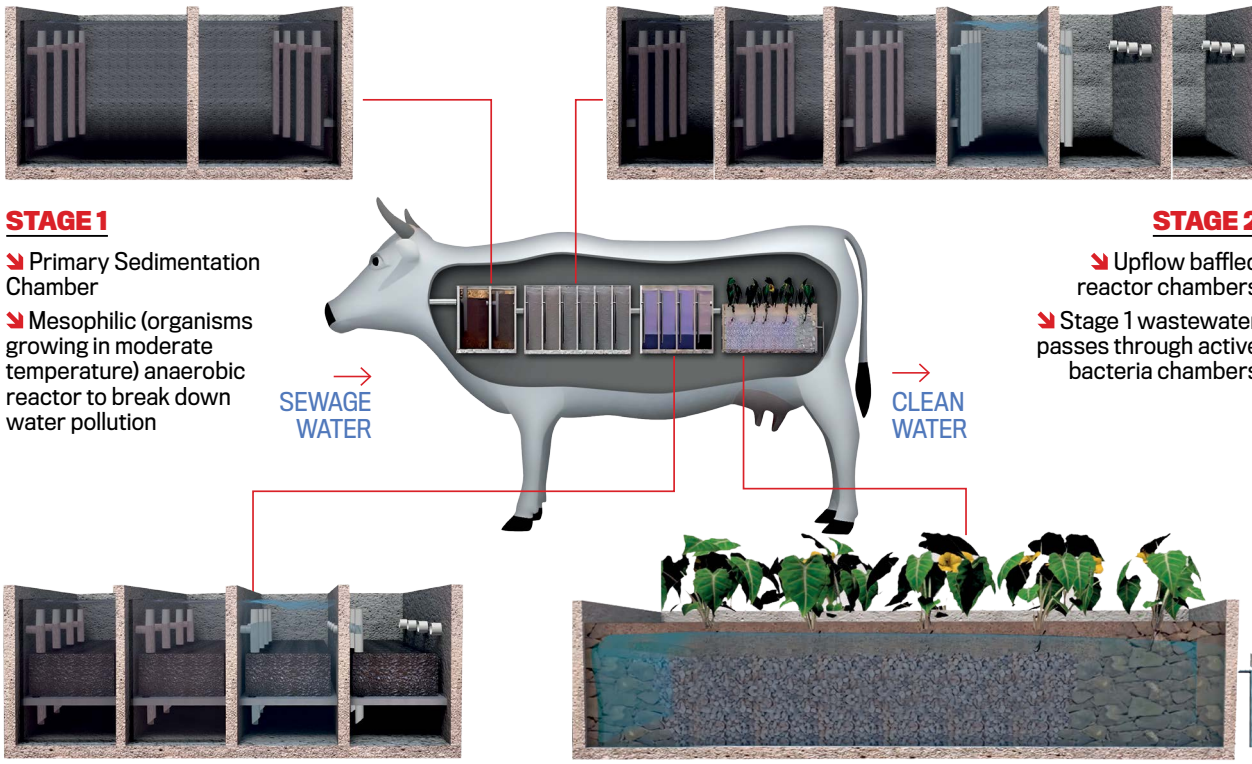
TIME FOR BOVINE **INTERVENTION**

Using biomimicry, ECOSTP utilises the principle of **multiple chambers and microorganisms found in a cow's stomach** to break down pollutants and uses gravity to move water



STAGE 1

- Primary Sedimentation Chamber
- Mesophilic (organisms growing in moderate temperature) anaerobic reactor to break down water pollution

STAGE 2

- Upflow baffled reactor chambers
- Stage 1 wastewater passes through active bacteria chambers

STAGE 3

- Attached growth biological filters with a high surface area
- Stage 2 wastewater passes through filter mass with bacteria lawn/ film

STAGE 4

- Stage 3 wastewater passes through a horizontal wetland with vascular plants and algal colonies
- Removal of suspended solids, pathogens and nutrients (nitrogen and phosphorus)

Graphic by TANMOY CHAKRABORTY

residential apartments and industrial plants where wastewater from toilets and kitchens is directed into underground septic tanks. On the bed of the tank is a layer of cow dung with a variant of anaerobic bacteria selected to break down faecal matter. As the sewage flows to stages two and three (*see graphic*), the bacteria do their job and the wastewater is filtered through gravel beds and sand. The last stage is a natural filter comprising a garden bed of gravel, where plants supply the recycled water with oxygen. The water finally obtained can be used for gardening or pumped up into overhead tanks to be used for flushing tanks.

"For every 100 litres of wastewater, 85 litres of clean water come out," says Kumar. Every two years, the sludge from the tank has to be removed, but the running costs, he says, are hardly anything. Currently, ECOSTP is working with IIT Jammu to study a variant of anaerobic bacteria that can break down chemicals. That can open the door for treating industrial effluents.

"My team and I measure ourselves by three metrics. How much bad water did we convert to good water? How much power did we save compared to a normal STP? And how much coal we saved, because 70 per cent of India's electricity comes from coal," says Kumar. "We are proud to have reclaimed 2 billion litres of sewage so far without power or chemicals." ■

HAPPINESS QUOTIENT

- **With ECOSTP Technologies, Tarun Kumar has processed two billion litres of sewage since 2017. For every 100 litres of wastewater, 85 litres of clean water is obtained**
- **Kumar has around 230 clients across India, including residential apartments and industrial plants where wastewater from toilets/ kitchens is directed into underground septic tanks**